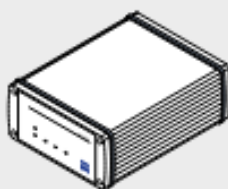
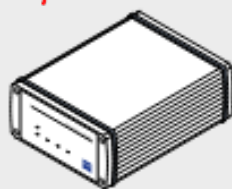


Stage-top Incubation

Temp

Gas side

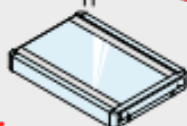


Gas Mixer System CO₂, ibidi
433617-9030-000

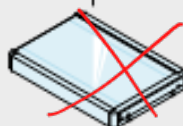
Gas Mixer System CO₂/O₂, ibidi
433617-9070-000

O₂ Gas Mixer Option, ibidi
433617-9040-000

Temperature Controller, ibidi
433617-9020-000



Glass Lid CO₂/O₂
heated, ibidi
433617-9000-000

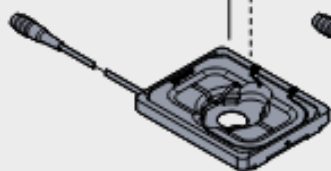


Glass Lid CO₂/O₂
unheated, ibidi
433617-9050-000

Heated glass!



Control Sensor T
Petri 60, steriliza
411857-9085-000



Heating Insert P S1
411860-9020-000



Insert for
Petri dish 35



Insert for
Petri dish 60

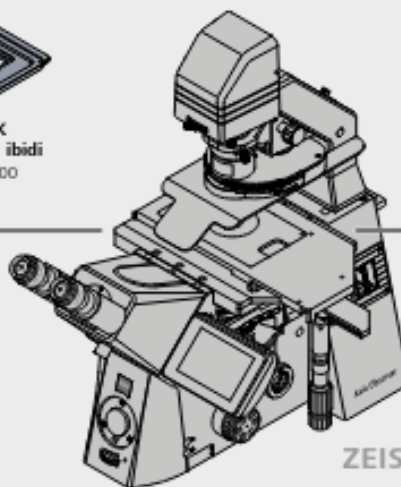


Insert for
slides and chambers



Heated plate K
with 3 inserts, ibidi
433617-9010-000

Separate
plate and lid
for multi-well
support



Stage for
K-frame inserts

ZEISS Axio Observer

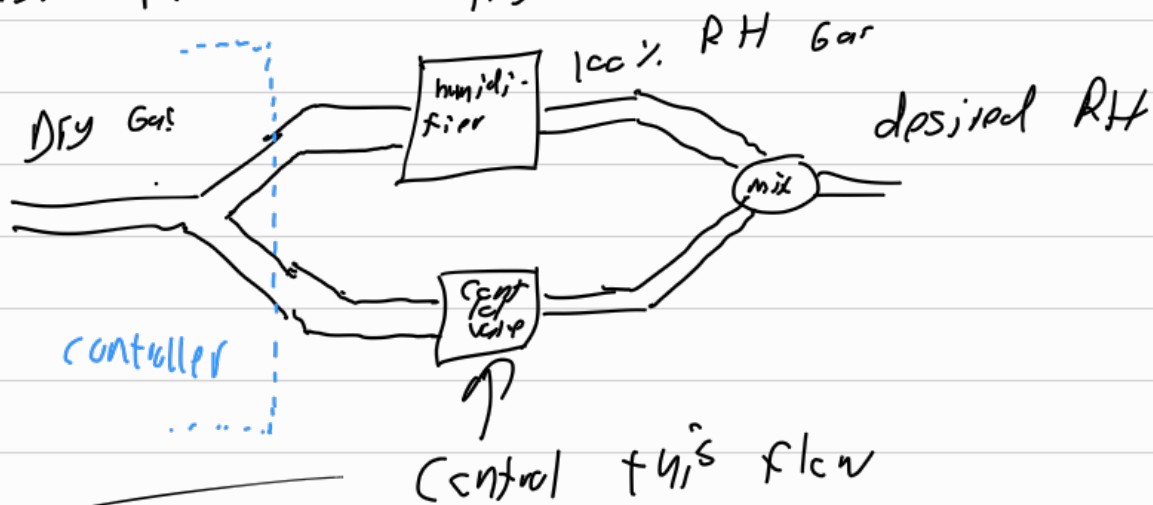
• Temp and CO_2 are maintained by a dual port system

Gas circulates around 2 units, bottom maintains temp, Top maintains CO_2 .

This Gas that circulates is then pumped from a mixing chamber to the incubator.

Gas is exhausted at the incubator *Not BSL3 safe! (unless filtered)*

Humidity is controlled by having a dry gas get mixed with 100% relative humidity gas to obtain the desired relative humidity



Separate holders for multi-well and slide/dish
only 2 though

Glass is heated directly with heating elements embedded in glass

Heating:

2 parts: 1. Temperature Controller

2. Incubation Chamber $\rightarrow$ Heated Plate ^{2 separate for}
 $\rightarrow$ Heated Lid ^{slide/dish and multi well}

Temperature Controller: Output per channel $\rightarrow$ Max DC 24V, 2A (45W)

Dimensions (mm): 90 x 170 x 230

Weight: 1.75 Kg

Heated lid: Dimensions (mm): 20 x 85.5 x 127.5
(slide/dish) Gas tubing: push-in fitting, OD 4mm

Range: Ambient + 5 $^{\circ}$ $\rightarrow$ 50 $^{\circ}$ C
(min 18 $^{\circ}$ C)

Accuracy: $\pm 0.2^{\circ}$ C (at sensor)

Uniformity: $\pm 5^{\circ}$ C (entire heated glass) ^{Not too important? since glass top is just for viewing, doesn't keep stuff warm}

Heated plate: Dimensions (mm): 15 x 85.5 x 127.5
(slide/dish) Observation Area: 40 mm x 82 mm

Range: Ambient + 5 $^{\circ}$ $\rightarrow$ 46 $^{\circ}$ C
(min 18 $^{\circ}$ C)

Accuracy: $\pm 0.2^{\circ}$ C (at sensor)

Uniformity: $\pm 0.5^{\circ}$ C (entire heated plate)

$\downarrow$
Heated plate fits into a multi-well mount on microscope stage

Heated plate has contacts to provide electrical connection to heated lid.

(previous connection to both temp and humidity sensor in the lid unit)

Heated lid is heated slightly higher than heated plate to prevent condensation from forming on the glass.

There are cut outs in the heated lid to provide tubing access for use with perfusion experiments. Cut-outs can be closed with a magnetic cover plate.

Slide holder uses magnets to hold the mount to the heated plate.

4.6.1 Slide Holder μ -Slide – Silver Line

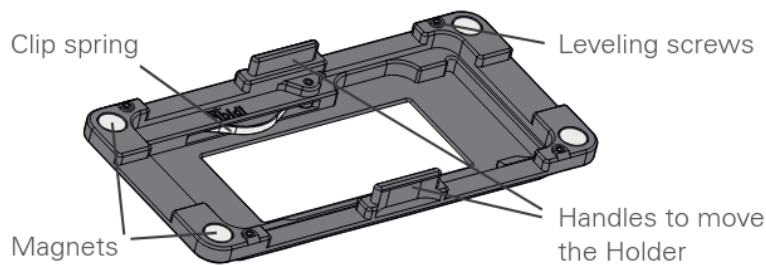
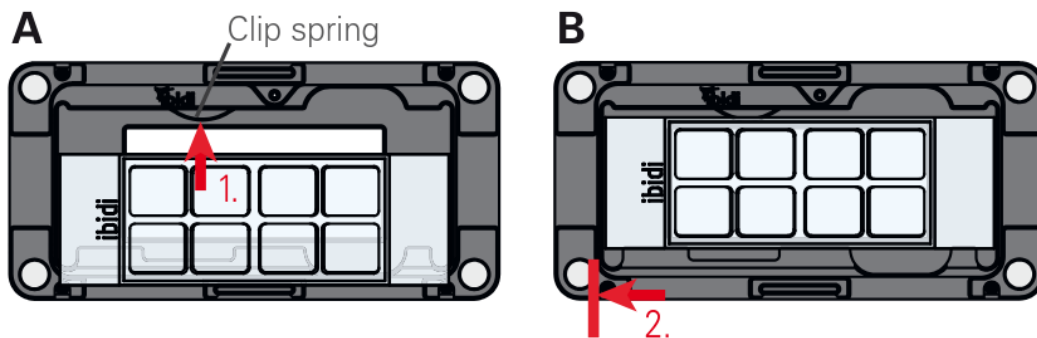


Figure 13: Slide Holder μ -Slide – Silver Line.

The Slide Holder μ -Slide – Silver Line comes as one piece, without top (Figure 13). The slides are held in position by a clip spring.



The slide is held in position using a clip spring.
The slide holder can be leveled using the leveling screws.

dish holder comes in two parts, the top and bottom.
↳ Bottom fits into heated plate,
top is used to press the sample down

Top can be flipped around to support "Tall" and "low" dishes

Top and bottom can be independently leveled using screws.

Top and bottom mount using magnets

Multi-Well System:

- Heated plate fits directly into K-type frame
- Heated plate requires channels 1/2 as well as 3 for heated plate and for heated Glass Bottom
- Supports calibration with sensor that makes direct contact with "dummy" media.

Gas Incubation System :

2 parts: 1. Gas mixer } uses heated tubing package
2. Humidifying column }

Gas mixer : Dimensions (mm) : $90 \times 170 \times 230$

Weight : 2.23 Kg

Gas tube Input : ID 4mm, OD 6mm (Polyurethane[®], hydrolysis resistant)
Gas tube output : ID 2.5mm, OD 4mm

Gas Input : Compressed air (1 bar)

: O_2 (Purity > 99%) (for PH)

Optional : Nitrogen (Purity > 99%) (for O_2 displacement)

CO_2 : Range : 0 - 15%

(1 bar) Accuracy : $\pm 0.2\%$ Vol + 3% of reading

Resolution : 0.1%

O_2 : Range : 1 - 21% (0.5% - 1% less accurate)

(1 bar) Accuracy : $\pm 0.2\%$ Vol (typical)

$\pm 0.7\%$ Vol (max)

Resolution : 0.1%

Humidity Control : Control range : 20 - 99% (lower limit dependent on env and gas supply)

Sensor accuracy : $\pm 2.5\%$

Resolution : 0.1%

Temperature control : $45^\circ - 55^\circ$

Temp Accuracy : $\pm 2^\circ C$

Temp resolution : 0.1%

Flow control: range: 0-200/h
Accuracy: $\pm 25\%$
Resolution: 0.1%

(flow control only
for CO_2/CO_2 version)

Humidifying Column: liquid: Ultra-pure water

min vol: 50ml

max vol: 190ml

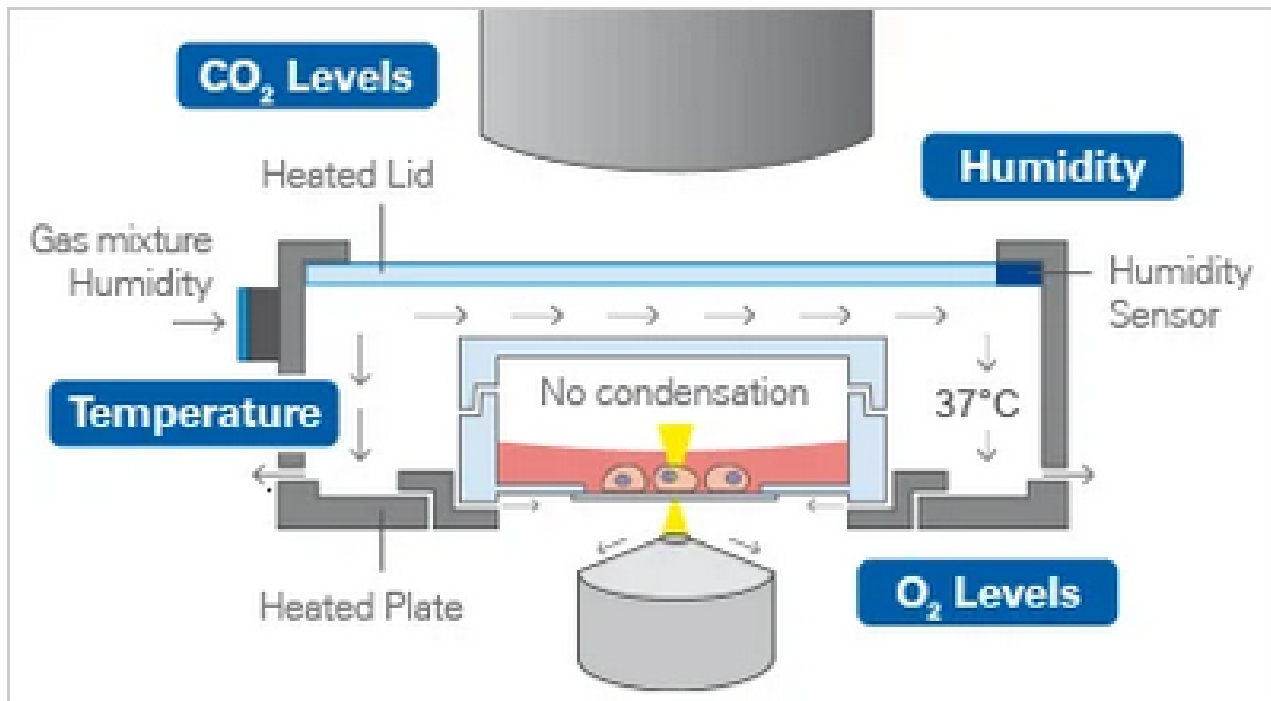
CO_2 unit only controls CO_2 concentration

CO_2/CO_2 unit can introduce O_2 and can displace O_2 by introducing N_2 . This can create hypoxia

- The Gas mixer adds controlled amounts of CO_2 to the Gas flow to maintain the set CO_2 concentration.
- The humidity of the Gas flow is measured by a humidity sensor located in the Incubation chamber.
- Humidity is controlled by adjusting mixture of "Dry Gas" flow and 100% RH Gas flow from the humidification tower
- Gas mixer requires pressurized Air for the air supply
- Gas is bubbled through heated water to get 100% RH water
- Humid Air tube is heated by a heating wire that feeds from the heating cable from the humidifying column
- For the slide/dish system, the humidity sensor is integrated into the lid and makes communication connection

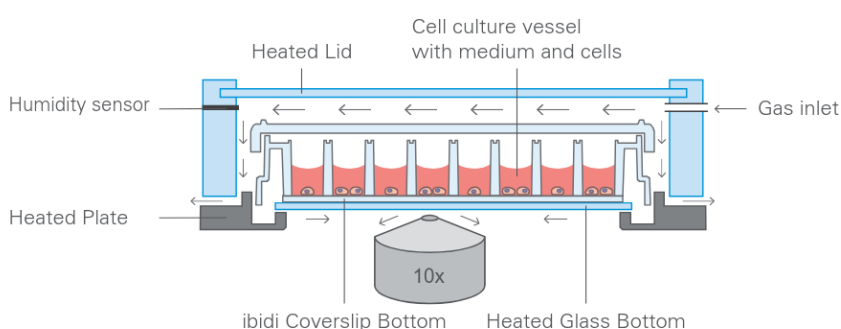
to the Gas mixer through the heated plate contacts as well as a bridge cable from the temperature controller

- For the multiwell system, the humidity sensor is separate and can be installed in the lid. This can be directly connected to the Gas mixer device.



Gas flow is designed to escape through the bottom of the Incubator.

- The CO₂ controller controls the temperature of the Gas flow



Gas flow designed to escape the Incubator

Conditions:

- Temp module 5 advises to avoid vibrations from Vacuum pumps, Centrifuges, electric motors, processing equipment, machine tools (may not be suitable for glass hood operation, why?)

- R recommends setting up Incubator with experimental conditions

Such as: air con, air flow, illumination, microscope settings, etc.

This indicates that the system is capable of operating in conditions that match fume hood in BSL3?

- Manual for Ibidi indicates that sample temp is supposed to be offset from the set values of the incubator. This is because there is a heat gradient that forms and so generally the incubator must be set slightly higher than the desired media temperature.

(Also generally, a vertical temp gradient is desired to prevent condensation.)

(heated glass is set much higher, than heated plate is set slightly higher)